MIL-D-50029B

15 JULY 1963

SUPERSEDING
MIL-D-50029A
18 JULY 1960

MILITARY SPECIFICATION

DYE, VAT YELLOW 4

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of dibenzpyrenequinone type dye which has a Color Index name of Vat Yellow 4.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 — Sieves, Standard for Testing Purposes

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20401. Orders for the above publications should cite "49 CFR 71-90".)

3. REQUIREMENTS

3.1 Purity and absorbency ratio. The purity of the yellow dye shall be no less than 80 percent and the absorbency ratio shall be no greater than 0.1 when tested as specified in 4.5.1.

FSC 6820

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3.2 **Volatile matter.** The volatile matter in the yellow dye shall not exceed 1.0 percent when tested as specified in 4.5.2.

3.3 **Particle size.** The particle size of the yellow dye shall conform to the requirements of table I when tested as specified in 4.5.3.

TABLE I. *Particle size*

Sieve size	Percent passing
840 microns (No. 20)	100 min
297 microns (No. 50)	97 min
44 microns (No. 325)	80 max

3.4 **Anti-dusting agent.** The yellow dye shall contain 2.0 ± 0.5 percent Bayol F (see 6.7) as an anti-dusting agent when tested as specified in 4.5.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the yellow dye offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.5).

4.4 **Sampling for test.** Sample containers shall be taken at random from each lot as follows:

Lot size	*Sample size*
1	1
2– 275	2
276– 545	3
546– 900	4
901–1345	5
1346–1875	6
1876–2500	7

Sample size for lots exceeding 2,500 containers shall be calculated using the following equation:

$$n = 0.15 \sqrt{N}$$

where: n = sample size and

N = lot size

A specimen shall be removed from each container in the sample and placed in a

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clean, dry container labeled to identify the lot and the container from which it was taken. Each specimen shall be tested as specified in 4.5.

4.5 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.5.1 *Purity.*

4.5.1.1 *Preparation of standard dye.* Place about 35 grams (g) of the dye in a 4-liter Erlenmeyer flask and add 2 liters of 1 percent sodium hydroxide solution. Heat to about 40°C. Reduce the dye by adding 10 to 12 g sodium hydrosulfite. Swirl to dissolve the vat soluble material. Allow the vat insoluble byproducts to settle. Rapidly filter the solution through a medium fritted glass funnel. (If a large amount of residue remains in the bottom of the flask, repeat the reducing process.) Discard the insoluble material. To the reduced dye solution, add 1 pint of 6 to 8 percent sodium hypochlorite for each liter of dye solution. Bubble oxygen or air through the dye solution to reoxidize the dye. (The reoxidized dye should be an orange color. If the color is brown, add more sodium hypochlorite and digest at 70° to 80°C. until the dye becomes orange.) After reoxidation is complete, continue to bubble air through the solution until the dye settles toward the bottom of the beaker. Decant the liquid. Centrifuge the dye solution to concentrate the dye. Filter through No. 41 Whatman paper in a Buchner funnel. Wash the dye by returning it to the beaker, adding distilled water, stirring the mixture to disperse the dye, and filtering. Repeat the washing process until the filtrate gives a negative test for free chlorine. After the free chlorine has been removed, test the filtrate with phenolphthalein. If necessary, continue the washing to remove any remaining sodium hydroxide. Drop the wet cake of dye into enough xylene to completely cover the cake. Heat the xylene to boiling to expel the water. (A Dean Stark extraction assembly is convenient for use.) Cool to room temperature. Filter through the Buchner funnel. Finally, dry in an oven at 70°C. for about 4 hours.

4.5.1.2 *Procedure.* Prepare the standard dye solution (see 6.4) and the sample solutions as follows: weigh to the nearest milligram (mg) 0.100 g of the dye and transfer quantitatively to a 400-milliliter (ml) beaker, using a small amount of sulfuric acid for washing. Dissolve in about 250 ml of sulfuric acid by means of magnetic agitation. Transfer to a 500-ml volumetric flask and dilute to the mark with sulfuric acid. Mix thoroughly. Remove a 10-ml aliquot and transfer to a 250-ml volumetric flask and dilute to the mark with sulfuric acid. Mix thoroughly. Transfer a portion of each solution to matched cells of a Beckman spectrophotometer and obtain the absorbency of each solution at 568 to 571 millimicrons ($m\mu$) and at 444 $m\mu$, using sulfuric acid in the reference cell.

$$\text{Percent purity of yellow dye} = \frac{100 \text{ abs (568 - 571 } m\mu\text{) unknown}}{\text{abs (568 - 571 } m\mu\text{) standard}}$$

where:

abs = absorbency

$$\text{Absorbency ratio} = \frac{\text{absorbency (444 } m\mu\text{) unknown}}{\text{absorbency (568 - 571 } m\mu\text{) unknown}}$$

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4.5.2 *Volatile matter.* Weigh to the nearest mg approximately 5 g of the specimen in a tared, glass-stoppered weighing bottle approximately 50 millimeters (mm) in diameter and 30mm high. Remove the stopper and dry to constant weight in an oven maintained at 70° to 75°C. Cool in a desiccator and weigh.

$$\text{Percent volatile matter} = \frac{(\text{Loss in weight})\ 100}{\text{Weight of specimen}}$$

4.5.3 *Particle size.* Sieves conforming to the requirements of Specification RR–S–366 shall be used for the particle size determination. Nest the sieves in order of decreasing mesh size, placing the largest mesh on top and a pan on the bottom. Weigh to the nearest decigram (dg), approximately 10 g of the specimen and place it on the No. 20 sieve. Brush the material with a camel's-hair brush until no more material passes through the sieve. Remove the sieve and weigh the material retained. Repeat this procedure with each of the other sieves. Calculate the percentage of material passing through each sieve.

Note. In case of dyes too fluffy to sieve dry, the following method of determining the particle size may be used: Weigh to the nearest dg, approximately 10 g of the material and mix in a beaker by stirring with a glass rod, with sufficient water to produce a smooth paste. Carefully wash the thoroughly wetted sample through the sieves. Dry the sieves in an oven at 70° to 75° C. and weigh the material remaining on each sieve. Calculate the percentage of material passing through each sieve.

4.5.4 *Anti-dusting agent.*

4.5.4.1 *Preparation of master solution.* Weigh 20 g of specimen to the nearest 0.001 g into a 250 ml beaker. Add 100 ml of carbon tetrachloride and agitate for at least one-half hour (specimen will not dissolve). Filter with suction. Wash the funnel and precipitate with 50 to 60 ml of carbon tetrachloride. Quantitatively transfer filtrate to a 200-ml volumetric flask and dilute to volume with carbon tetrachloride.

4.5.4.2 *Test procedure.* Obtain the infrared spectrum of the specimen in the range of 2.1 to 2.6 microns using a Cary Model 14 Spectrophotometer having 10.0 centimeter cells.

(*a*) *Operating conditions.* Caution: Be sure coolant water is on and is flowing freely through the instrument and lamp.

Power Switch	ON
Selector Switch	IR
Slit Control	15
Slit Height	IN
I R Detector	IN
Phototube Selector	IN
Source	VIS or UV
Marker Switch	100A
Pen Switch	ON
Slit Switch	ON
Chart Switch	OFF
Range Selector	AUTO
Speed Control	X20
Scan Switch	OFF
Master Switch	OFF
I R Source	ON

(*b*) *Operating procedure.* Place the solvent used or the prepared blank solution in both sample and reference cells of the proper length. Wipe the windows thoroughly and place one cell in each cell compartment.

Set the baseline as follows:

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Turn "master switch" to ON. Using the "Scan Switch," set the wavelength to the upper limit of the desired spectral range. (The wavelength setting will increase or decrease as the "Scan Switch" is put in the + or — position, respectively). By means of the "Balance Control" set the pen at Optical Density 0.0.

Set the instrument to the highest wavelength in the desired range for which there is a balancing "Multipot." Adjust baseline to 0.0 by means of the "Multipot" for that wavelength. Set instrument for next lowest wavelength for which there is a balancing "Multipot." Continue until the lower limit of the spectral range is reached.

Set instrument to upper wavelength limit and turn "Master Switch" OFF. Turn the "Chart Switch" to ON, turn the "Scan Switch" to —, turn the "Master Switch" to ON, and record the baseline. If deviations from 0.0 are observed, make corrections with the "Multipot" using the above procedure. When the baseline has little or no deviation from 0.0, set to the upper wavelength limit of the desired spectral range, and then turn the "Master Switch" to OFF.

(c) *Test procedure.* After the baseline has been set, remove the cell from the sample compartment. Rinse thoroughly and fill with the solution to be tested. Wipe the windows to remove any spilled solution or finger prints, etc. Place cell in sample compartment and replace the cover. Turn the "Scan Switch" to — , turn the "Chart Switch" to ON, turn the "Master Switch" to ON, and record the curve through the desired range. When complete return the "Master Switch" to OFF.

4.5.4.3 *Calculations.* Determine the optical density at the maximum occurring at about 2.31 microns. The percent Bayol is determined from the calibration curve (fig. 1).

4.6 **Acceptance/rejection criteria.** If any specimen fails to meet the test requirements of this specification, the lot of represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 **Packing, level C.** The material shall be packed in steel or fiber drums conforming to Specification 37A or 21A, respectively, of the Interstate Commerce Commission Regulations for transporting explosives and other dangerous articles. Each drum shall be provided with a minimum 0.004-inch thick polyethylene bag liner which shall be closed by knotting, tying, or heat sealing.

5.2 **Marking.** In addition to any marking required by the contract or order, shipments shall be marked in accordance with Standard MIL–STD–129.

5.2.1 *Precautionary marking.* Each container shall be conspicuously labeled:

"CAUTION: Avoid skin contact or breathing of dust or fumes; use with adequate ventilation."

6. NOTES

6.1 **Intended use.** This specification covers a dye which is intended to be used in the manufacture of colored smoke mixtures.

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6.2 **Ordering data.** Procurement documents should specify the title, number, and date of this specification.

6.3 **Nomenclature.** A commercial name for this dye is Indanthrene Golden Yellow GK. It is listed in the Color Index under the number CI 59100.

6.4 **Standard samples.** Five g of a reference standard sample for use in testing purity in accordance with 4.5.1.2 may be obtained from:

Director for Quality Assurance
ATTN: Laboratory Facilities Branch
Edgewood Arsenal, Md.

6.5 **Batch.** A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing process intended to make the final product substantially uniform.

6.6 **Significant places.** For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.7 **Anti-dusting agent.** Bayol F is a highly refined petroleum oil obtainable from Standard Oil Company of New Jersey.

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Army—MU
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Preparing activity:
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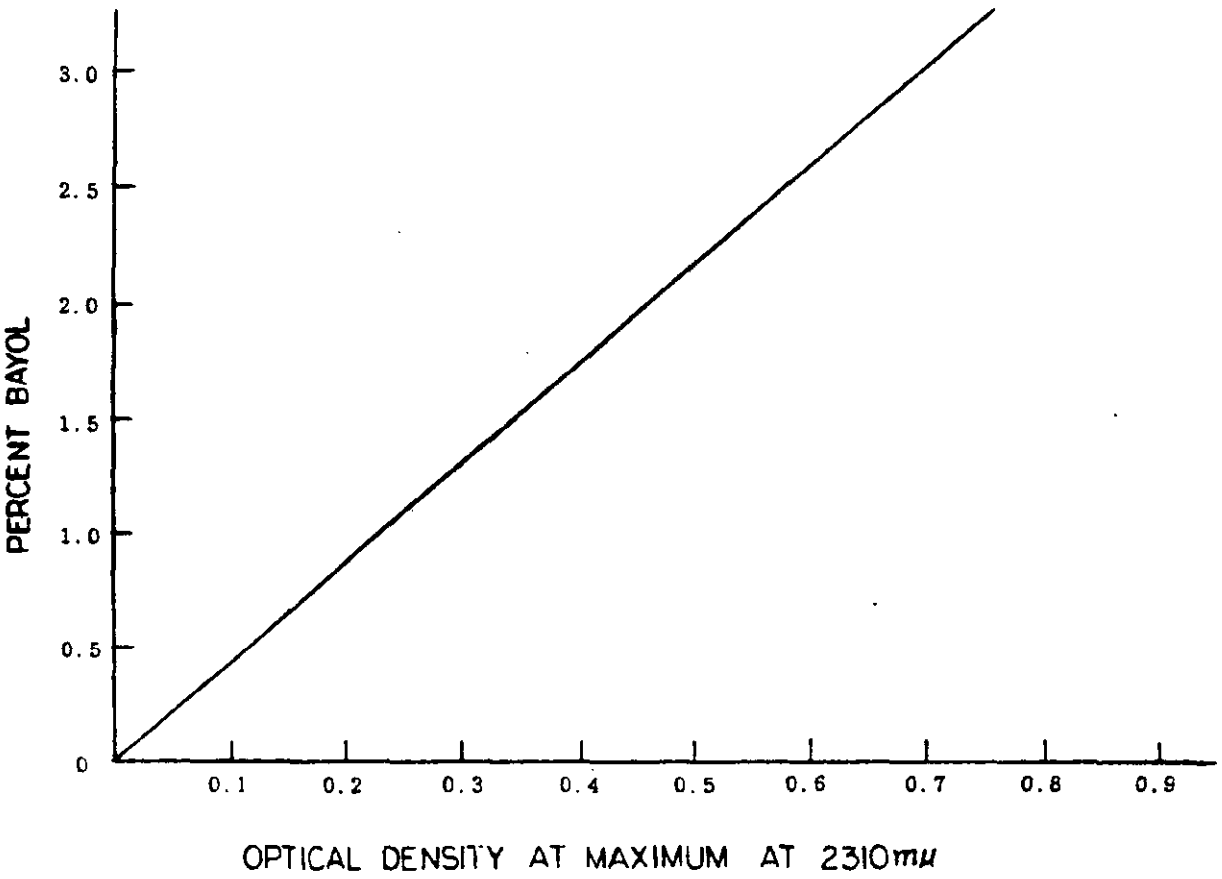


FIGURE 1. *Calibration curve dye, oil yellow 4.*

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